A Galaxy Law Tied to Dark Energy, and a Map of What It Leaves Unexplained

Working note (PAPER67), version 1.1, 2026-11-02. AI-assisted research programme; not peer reviewed. $\kappa = \frac{1}{2}$ is fitted. The theory is not closed. Nothing here says the data favour this framework over any other.

Abstract

We report what is left of a research programme after its own final grade. A galaxy law, $g_N = g^2/(g + a_0)$ with $a_0 = \kappa c\sqrt{G\rho_{\rm DE}}$ and $\kappa = \frac{1}{2}$ fitted, describes isolated disc galaxies well on the canonical footing: SPARC rms 0.0541 dex at median Υ 0.52, no tilt with speed, no slope with surface brightness at fixed speed. The law is a restatement of MOND's simple form with a proposed origin for its scale. Earlier authors proposed that origin, and the scale's tie to dark energy, as distinct from cH_0, has no decisive test yet.

The law fails IC 1613, several Local Group classes, cluster shear and X-COP, whose hydrostatic masses it misses by a factor of 2.93, and the unscreened Solar System. Its Milky Way result depends on a vertical form it doesn't supply, and at 60–150 $\mathrm{km\,s^{-1}}$ it asks for about twice the stellar mass estimated from photometry. It has no action and supplies no dust-like term for the CMB. To cover those we developed a cold dark energy sector. On measured data every graded candidate fails, including the law alone, and the sector is now retired. We publish the scorecard, the two walls that stopped it, and twelve requirements any successor must meet. This is numerical evidence for the families tried, not a no-go theorem. We close with dated tests that can still kill the law.

Version 1.1 adds lanes run after the retirement (Section 7). None was blind, all were run by AI subagents, and no outside group has reproduced them. A gas term fitted on X-COP reaches 14 of 15 keys on the canonical footing, fragile, never 15, and tied to present gas it is dead on the Bullet. A repaired test gives no statistical support that the a_0 formula is special; that doesn't refute it, and the test lacks power. A line of v1.0 about v9's α_1 is corrected. A published rise of a_0 with redshift (MUSE-DARK III), which v1.0 omitted, counts against the law's flat band at face value; we read it as mass-route dependent and not decisive, and we say what that reading rests on. Formula searches that have finished found no hit, and two levels keep running with no result claimed. A definition audit moves single lines of the scorecard, and each candidate still fails at least two on every reading run.

1 Scope

This is a programme report, not a complete theory. Three owner decisions on the record set its form: the a_0 floor was removed from the cold sector (2026-10-31), the search for a derived threshold stopped (2026-11-01), and the cold sector was retired (2026-11-01). Earlier deposits stand as dated records. The two footings, canonical $a_0 = 9.3603 \times 10^{-11}$ and alternative $1.1312 \times 10^{-10}\,\mathrm{m\,s^{-2}}$, are never pooled. This paper's audit script checks the quoted numbers against committed files. It doesn't check the prose.

2 The law, and whose idea it is

$$g_N = \frac{g^2}{g + a_0}, \qquad a_0(z) = \kappa c\sqrt{G\rho_{\rm DE}(z)}, \qquad \kappa = \tfrac{1}{2}\ \text{(fitted)}.$$

Credit. The first relation is Milgrom's modified dynamics in its simple form [1]. The numerical nearness of a_0 to cH_0 and to $c^2\sqrt{\Lambda}$ is his observation [1, 2]. Limbach, Psaltis & Özel [3] proposed the tie to the dark energy density. None of this originated here.

κ isn't derived. Of 537 attempts on the record, none derives $\kappa = \frac{1}{2}$. It isn't singled out by SPARC either. A profile likelihood over 175 galaxies has a free best fit of 1.0766×10^{-10}. Against it, $\kappa = \frac{1}{2}$ on $\rho_{\rm DE}$ reads $\Delta\chi^2$ 63.90 (1.82σ with galaxy-clustered errors), the alternative footing 7.04 (0.60σ), and Milgrom's own $cH_0/2\pi$ 5.34 (0.53σ). So Milgrom's value fits at least as well as ours.

What this programme adds: a reconstruction of $a_0(z)$ from raw BAO and supernovae with a stated ruler, the graded comparison below, and the failure record.

The band, and what it rests on. The reconstruction gives $a_0(z)/a_0(0)$ of 0.91 [0.76, 1.03] at $z = 1$, 0.94 [0.71, 1.12] at $z = 1.5$ and 0.91 [0.33, 1.21] at $z = 2.33$ (68 %), against 1.76, 2.35 and 3.49 for the rival $a_0 \propto H(z)$. Constant dark energy isn't rejected (1.97σ, primary). **This band isn't free of the retired sector's input:** it

takes $\rho_m(z) = \omega_m(1+z)^3$ with a cold mass from the CMB, and with positivity alone, no CMB and no H_0, its check fails. The law doesn't explain that matter term (Section 5).

The distinctive claim has no decisive test. At $z \approx 0$ the law can't be told from $a_0 \propto cH_0$. Separating them at 3σ needs a_0 to 0.055 dex at $z = 1$ and 0.084 dex at $z = 2.33$. The best public sample reaches 0.4–0.5 dex.

A published rise that counts against the flat band at face value. MUSE-DARK III [13] fits the radial acceleration relation in 79 galaxies at $0.33 < z < 1.44$ and reports $a_0(z \sim 1) = 2.38^{+0.12}_{-0.10} \times 10^{-10}\,\mathrm{m\,s^{-2}}$ and, for $a_0(z) = a_0(0) + a_1 z$, $a_1 = 1.59 \pm 0.10$ in the same units (95 % intervals). Our band at $z = 1$ on the canonical footing is $0.91 \times 9.3603 \times 10^{-11}$, a factor of about 2.8 lower. Alcock [14], an independent author, fits the survey's four bins as read from its figure: a constant a_0 is excluded at 5.3σ if the bars are 1σ and at $\geq 10\sigma$ if they are 95 % intervals, the reading the survey's stated convention for its main fits points to, statistical errors only. On that reading $a_0 \propto E(z)$ is itself in tension (4.2σ). The Letter notes that frameworks tying a_0 to the cosmological constant are under the same pressure. That is this law. It was received 2026-07-10, before our first deposit using these data (2026-09-06), and we claim no priority on it.

Why we read it as non-diagnostic, not as a pass. The survey's main fit leaves stellar mass free of its SED estimate. Our own frozen run (CFG262, a different sample of 109 galaxies, not a direct a_0 measurement) finds the rise travels with the mass route: the implied change from z 0.52 to 1.20 is $+0.570 \pm 0.094$ dex on fitted masses, -0.115 ± 0.489 on SED plus H_2, and $+0.036 \pm 0.138$ on SED alone, with a spread of $\times 12.7$ in level in the top third. The SED routes are consistent with flat and also with $H(z)$, so they rescue nothing. A redshift-dependent mass drift of 0.13–0.19 dex across the bins would remove the rise; the survey puts its unmodelled molecular gas near 0.2 dex, but nothing here shows that drift is present. MUSE-DARK II [15] finds no baryonic Tully–Fisher zero-point evolution at $z \sim 1$ (0.00 ± 0.06 dex), with a further ± 0.16 dex from the local zero point in its own model comparison. On the survey's masses, the band and the dark energy tie fail. That reading stands until a per-galaxy reanalysis with independent baryon masses says otherwise.

3 What it does in isolated galaxies

Table 1: The law on isolated systems. FG1 is the frozen final grade; a clean-room regrade (CRG1) reproduced its table, 289 match, 6 close, 0 disagree, mainly a test of code.

line	canonical	alternative	lane
SPARC rms, median Υ (line 0.50–0.80)	0.0541, 0.52	0.0537, 0.44 **fails the line**	FG1
tilt with speed	+0.02	−0.02	FG1
LITTLE THINGS, median E	−0.025	−0.064	FG1
63 groups, Z	+0.31	−0.31	FG1
slope with surface brightness, 60–120 km s^{-1}	-0.016 ± 0.061, flat	-0.023 ± 0.063, flat	HS1
local a_0, MeerKAT against SPARC	+0.032 dex (line 0.15)		PAPER60

A caution on the Υ line. The 0.50–0.80 line was first written for this law, and its lower edge is a population model's central value, not a measurement. So passing it is weaker than it looks; Section 4 sets it against measured masses.

Surface brightness is the strongest result. At fixed speed, a halo set by mass alone can't follow the disc: every version of the retired sector's halo tilts, −0.17 to −0.29 per dex, and the law doesn't. That's a real discriminator on the data used, between the law and these halos only.

Isolated-galaxy lensing is shape, not a clean amplitude. KiDS-1000 prefers the law's shape over the retired halos ($\Delta\chi^2$ +366.0 against the best native halo). But the law's amplitude depends on colour and on the baryon inventory: blue lenses read $A = 1.016 \pm 0.086$ in one lane and 0.878 in a later forward model that adds a hot gas inventory, and red lenses 1.587 and 1.313. No tested model fits these profiles in absolute terms.

4 Where the law fails

- **Milky Way.** The law gives only the in-plane force, so the vertical test needs a form it doesn't supply. The field-equation form (AQUAL) is INFEASIBLE, 0 cells, on both footings. So is a real QUMOND solve, which differs from AQUAL by at most 0.13 % in $K(R_0)$, and varying heights and thick-disc share doesn't reach the line: in priors the lowest is 75.9 (alt 76.2) against its top of 70.8 (UX5, K_z FAILURE SURVIVES). That rests on a kinematic summary fitted inside Newtonian potential families. The round rule, which has no field equation behind it, passes in 1 in-prior cell (alt 6) (LW1), 72 (alt 196) with a shorter thick disc, and there it still misses the K_z scale length. On an older, weaker gate, 7 of 175 cells pass (alt 4), all with a short disc and the lightest local column; those have $K_z(R_0)$ 76.7–80.9 against 66.4 ± 2.2, +4.7 to $+6.6\sigma$, and leave the outer slope excluded (Z −3.88). FG1 counts this as FAIL.
- **IC 1613.** $E = -0.593$, $Z = -7.9$; it fails under every halo-free inclination. One galaxy: disfavoured, not excluded. $Z = -7.9$ has no velocity error; with 15 % on V the base is −3.94, and −2.35 and −2.83 at RF2's central inclinations 34.4 and 37.9. Only inclination could close it: $E = 0$ needs i 22.1 (alt 21.1). A disturbed gas disc is a live explanation, not computed (UX4).
- **Local Group classes.** Z: MW ultra-faints +5.22, bright M31 +2.20, other M31 +2.29, MW classicals +1.69, field +0.57. Adding the external field makes it worse: ultra-faints +6.76, bright M31 +3.97. And at fixed M_V the data are flat in host field, -0.05 ± 0.19 dex per dex, where AQUAL with its external field predicts −0.76, so neither version describes these satellites (LW1). Listed systematics don't close them (UX4, FAILURES STAND, both footings). A central stack of binaries, tides, membership and heating leaves ultra-faints at Z +5.08 and other M31 at +2.75; bright M31 reaches +1.47 only with an assumed orbit nobody measured. A ceiling stack overshoots and breaks what passes, field Z −7.67. The field's pass doesn't test this regime: 41 satellites are fainter than the faintest field dwarf with a dispersion.
- **Measured stellar masses, 60–150 km s^{-1}.** The law's Υ is 0.62 and 0.65 where GSWLC stellar masses give 0.277 and 0.304 (Z +3.4, +5.6). These are estimators against W1 luminosity, not a 3.6 μm measurement. Gas dominance doesn't excuse the 100–150 bin. Offsets, distances and inclinations don't carry it (UX5, NOT CARRIED). The level miss against GSWLC is narrow, best $|Z|$ 2.12 at +0.204 dex; Taylor needs more than a Salpeter shift; and no offset changes the step between 100–150 and 150–200, which rests on 23 and 25 galaxies. Galaxies with accurate distances still read 0.622±0.063.
- **Disc weighings.** On this law's own form, DiskMass σ_z reads round rule $r = 1.239$ (alt 1.204), outside 0.90–1.10, a FAIL as frozen (LW1). Earlier lanes read $r = 1.250$ and 1.342 with a different interpolating function. Same-population dispersions [7, 8, 9] correct r by c of 1.414–1.517, giving 0.824–0.884 one-zone and, with exact Jeans and measured thickness, inside the line. But the sources' own tables run c from about 1.24 at the relevant radius to 1.88 in integrated light, so the question is open (OD1, NOT SEPARATED). NGC 628 reads $\Upsilon_{3.6} = 0.75 \pm 0.18$ and NGC 6946 about 0.4.
- **The Solar System.** The unscreened law fails Cassini's Q_2 by 2.44–4.59 times the 2σ line (alt 2.85–5.19). The pass in earlier reports belongs to a screened covariant theory. That line's final verdict (v3s) was that it holds as a galaxy law and fails as gravity plus cosmology. So the law as written here needs a screen it doesn't contain.

5 What it leaves unexplained

- **Cluster cores and shear.** From measured gas, in 318 eROSITA clusters with raw KiDS shear, the law is short by $A = 1.91 \pm 0.12$ at 0.5–1.2 Mpc (alt 1.75 ± 0.11) and matches at 1.2–3 Mpc, $A = 1.01 \pm 0.09$. On the measured chain it fails 10 of 15 keys on both footings, G-OUT-M3 +4.76, joint p 3e-8. On this statistic that is worse than the retired native profile (LW1).
- **X-COP.** The law's own number is +0.467, rms 0.476: $M_{\rm HSE}/M_{\rm law}$ of 2.93 (alt 2.72), all 7 clusters short. Taking 0.12 dex off the hydrostatic masses still leaves 2.23 (alt 2.06). An earlier scorecard showed another rule's row here; LW1 corrects it.
- **What is missing in clusters, as a description.** Inside about R_{500} the deficit is near 8 times the gas at 0.2–0.5 R_{500} and 5 times at 0.5–1 R_{500}, with no detected mass dependence, no BCG link and no preferred scale found, picked from 31 looks (LM1). Trained on KiDS, it predicts X-COP's hydrostatic masses out of sample, but HSC lensing reads it about 1.7 times higher, and tracking gas can't be told from tracking M_{500}, so its frozen label is NOT (GM1, which issued labels after a failed check and says so). No mechanism follows.
- **Outer shear against gas mass.** $\Delta\Sigma_{\rm OUTER} \propto M_{\rm gas}^{s}$ with $s = 0.90 \pm 0.21$, against the law's 0.47 and Newton's

0.84. The amplitude agreement above is an average over a slope the law misses.

- **The CMB and growth.** a_0 is absent from the linear FRW equations, so the law can't act there. The record has no CMB pass without a dust-like component. With one, a window exists at $z_{\rm on} \in [3 \times 10^4, 10^6]$ on the channels run. Retiring the sector leaves this unanswered. **The law offers no replacement, and we don't claim it needs none.**
- **The matter term.** The law takes $\rho_{\rm DE}$ from an expansion history whose matter term is several times the baryons. That term is unexplained here.
- **An action.** None is selected, and the a_0 sector's action stays open. Attempts on record are this programme's own versions on AeST [4] and Khronon [5], and don't test those theories as published. **Correction to v1.0:** v1.0 said v9, an AeST embedding, fails the preferred-frame bounds with $\alpha_1 = -2(K_B + 2)$. That is a static-branch ratio. A source moving through the aether has $\alpha_1 = \alpha_2 = 0$ at $G = \tilde{G}$ (AS3). v9 stays non-viable, because it then loses the law for moving systems. v3, Khronon with an added term, fails linear growth: the published cap $\mu_K \leq 14.8\,h/\mathrm{Mpc}$ sits 24–172 times below what σ_8 needs. v4, AeST with that term, is ill-posed where the aether twists and runs away on FRW for the dust shape tried. v3s, v3 with switched stiffness, is frozen as a galaxy law that fails cluster cores, linear growth, hosted dwarfs and the Milky Way disc.

6 The cold sector: tried and retired

What it was. A gravitating component with, at linear order, $w = 0$, $c_s = 0$, geodesic motion and an abundance set by an integration constant. A referee will read those as the linear behaviour of cold dark matter, and that's correct; it was never proposed as a particle, and the programme's no-particle rule is unchanged. Its candidates: C1, the native collapse rule N; C2, N with the measured gravitational response; C3, a basin split at $T \approx 10^{12.3}\,M_\odot$; C5, the native shape with the measured response; with C4, the law alone, as the control.

Table 2: FG1, canonical, with C4's X and DM from LW1 and C5 from HM1. P pass, F fail, – not graded, which isn't a pass. No winner is declared, because the lines aren't independent. Other readings of a definition move single cells (DA1, Section 7); the table stays as frozen.

	S	Tl	LT	G	K	X	Ln	Dw	MW	DM	IC	alt
C1 native N	P	F	F	P	F	P	F	F	F	F	F	no a_0, same
C2 N + response	F	F	F	P	F	F	F	–	F	P	–	no a_0, same
C3 basin split	P	P	P	P	F	P	P	P	F	–	F	also S, Dw fail
C4 law alone	P	P	P	P	F	F	P	F	F	F	F	also S fails
C5 native + response	F	F	F	F	F	F	F	F	P	P	F	not listed

S SPARC, Tl tilt, LT LITTLE THINGS, G groups, K 15 cluster keys, X X-COP, Ln KiDS isolated, Dw Local Group, MW Milky Way, DM DiskMass, IC IC 1613. C4's MW cell depends on the vertical form. C5's row is a reading of its numbers: HM1 printed NO VERDICT as frozen because one mass check failed, and an amendment moves no key by over 0.001. FG1's tag [GAS-CONDITIONAL]: at X-COP's gas scale +0.111 the native rule reaches 15 of 15 keys, rescuing K only; the measured scale is $+0.005 \pm 0.017$.

Wall 1: spirals at 150–250 km s^{-1}. Whatever is dust in the CMB falls into galaxies, and a live disc measures it contracting: $f_Q = 0.872$ [0.54, 1.07] (LC1), 0.946 [0.78, 1.08] with the disc grown from turnaround (PZ1). The SPARC side needs net f of about 0–0.2 at the 0.50 Υ line (RM2, six of 75 cells, three fitted numbers). This is the known Tully–Fisher zero-point problem [6] in measured numbers. A cored halo's response is somewhat weaker, $f_B = 0.753$ [0.729, 0.777] (DR8), not enough. Whether the 0.50 line is itself too high is open (Section 4, UL1, OD1); moving it wouldn't clear tilt or LITTLE THINGS.

Wall 2: outer cluster shear on measured gas. G-OUT-M3 reads 1.56 against 1.04, $Z = +4.41$ statistical, about +3.03 to +3.51 with systematics (M3A1). It's not an artefact on anything measurable from disk, nor the model's edge, and native two-halo supplies 3–11 % of +21 to +27 %.

No derived switch. The basin split's T was located, never derived. Three frozen attempts failed: 13.94 for the turned-around mass and 13.54–13.89 for a clock, against a window of 12.19–12.39 in $\log M$, and a candidate action had no link to mass. In shift-symmetric $P(X)$, a turn-on has $c_s^2 < 0$ at all 14 grid points, a result within that family (BOLTZ1).

Gas retention: the mass method decides it, and it isn't settled. A gas-fraction line from hydrostatic

Table 3: Doors tried in the closing push, by each lane's own label. Full list: the closing ledger.

idea	lane	result
density term in the law	DR1	SPARC BREAKS, CLUSTERS NOT REACHED
phase-space cap on response	DR7	BITES, FAILS [NOISE-LIMITED]
retention replaces the mass line	RET1	DERIVED LINE FAILS, FITTED ϵ FAILS
barotropic halo	HB1	MOVES, FAILS
fresh hunt, 19 routes	MH1	no mechanism survives whole
starting point	SP3	no response-on start reaches Υ's line in the owner's grid
symbolic search	SL1	TRAIN ONLY, 0 survivors
exact actions per object	AR1	MOVES AND FAILS

masses, fitted without X-COP or shear, takes the native profile from 12 to 14 of 15 keys with X-COP passing, G-OUT-M3 +3.48, +3.09, with frozen label NOT RETENTION by one cell (FR1). Okabe's HSC masses on eRASS1 gas read 0.216 dex below that line, -4.4σ; through the same grader that overshoots, 4 of 15 keys, G-OUT-M3 −2.54, X-COP failing. A second HSC reading, Umetsu on XXL gas, sits on the hydrostatic line instead (FR3). Neither line passes all 15 keys and X-COP. Both lanes were frozen after the retirement, weren't blind and don't touch galaxies.

Status. Retired by the owner on 2026-11-01. One lane frozen before that decision, a per-galaxy action response (AR1), landed after v1.0's deposit. Exact isotropic actions at each object's own halo and baryons give Υ 0.45/0.46 against Blumenthal's 0.36/0.37, on two backgrounds, so the line stays missed. The lift goes to the wrong bins: tilt +0.22/+0.23, LITTLE THINGS +0.129/+0.122. Keys read 14 of 15 on gas +0.005, X-COP −0.103, rms 0.150, groups +1.50. There is no mass trend in the response. The solver follows live runs at 10^{10} and $10^{13}\,M_\odot$, where Blumenthal over-predicts. The frozen label, MOVES AND FAILS, is conditional: check K1c missed its line, 2.8e-5 against 1e-6, not traced, and K2 and K7 first failed and were fixed by finer grids. No later result can reinstate the sector; only the owner's decision can. What survives from this work as fact, not as a theory: native collapse passes X-COP on hydrostatic masses, +0.013, rms 0.075, and one adiabatic response has the opposite sign in discs and cluster cores.

7 Lanes after the retirement

Read these knowing how they were made. Every lane in this section was frozen after the retirement, on 2026-11-02. None was blind: each read the record first, and several wrote down what they expected. All were run by AI subagents, and no outside group has reproduced any of them. They reinstate nothing. The one replication below (CU1) came from the same chain and mainly tests code. BF1's level L4 and BK2's level L2 are still running, and nothing here claims a result from either.

A gas term on clusters (UX3, CU1). The rule $g = \text{law} \times (1 + \theta\, k\, g_{\rm gas}/a_0)$, with k 13.29 (alt 14.25) fitted on 4 X-COP clusters, takes canonical keys from 10 to 14 of 15, G-OUT-M3 +0.58, +0.35, X-COP −0.016, rms 0.089. G-OUT-all still fails, −2.71, −2.77. Alt reads 10 of 15. At $k = \pi^2$, keys read 12 of 15 and X-COP fails. Unswitched, it breaks discs. A switch that shuts it off in discs passes largely by construction, and can't be told from a "clusters are big" control. There is no action or field equation, and the frozen label is KEYS FAIL. CU1 rebuilt it (73 match, 4 close, 1 disagree of 78) and stressed it. The 14 is fragile: it holds only for k 12.87 to 15.09, and k fitted on all 7 clusters reads 13. Joint p falls from 0.38 to 0.18 at the worse h (alt 0.10). G-OUT-all needs $k \leq 7.2$, G-OUT-M3 $k \geq 8.8$, G-MID-T3 $k \geq 12.9$, so no k gives 15 of 15, and no bootstrap draw does. If hydrostatic masses are 0.12 dex high, k is 8.38 and keys 11 of 15.

That term in mergers (UX6, MG1). On the Bullet's map [10], all seven readings read DEAD, both footings, both k. Local forms supply about the right mass, M_{300} 4.92, 2.44 against 3.5, 2.3 ($10^{14}\,M_\odot$), in the wrong place: $\Delta\kappa$ at the subcluster goes from −0.111 on the law to −1.158. Observed peaks sit on the galaxies. Only gas memory carried with the galaxies gets the sign right, and it fails on the main peak and on mass, and owes a carrier nobody has named. MG1, PROVISIONAL, from published numbers read through a web summariser: 8 of 9 subclusters that can decide have peaks on galaxies, 0 on gas, 1 between, in 6 systems. The sample was chosen for visible offsets, so that isn't a rate. Another reading of one criterion gives 7 of 12, and rows with published errors, not at the edge, number 4 in 3 systems. Abell 520 is contested. It isn't a map-level test.

A time lag in that term (TM2). Memory at past gas positions in a fixed frame reads DEAD at every lag on the Bullet, both footings. Vector memory carried along galaxy worldlines gets peaks and sign, misses the main mass at fitted k (Z +4.03, +3.89; alt +3.77, +3.63), and passes only post-result, at a fitted lag of 300–600 Myr and k of 7–10 against X-COP's 13.29, with 7 the edge of the scan. Stored per star, orbits erase it and relaxed clusters fail. Stored on the mean flow, discs spin up in a median 1.4–2.7 Gyr, an estimate on rotation-curve points. A third reading stays open and owes the k gap and an action. Its check K1 failed as frozen, so by its own rule these labels aren't a clean verdict.

k itself (KS1). $(M_{\rm tot} - M_{\rm law})/M_{\rm gas}$ is NOT ONE NUMBER, χ^2 473/14 (alt 528/14). It falls with radius, slope -0.52 ± 0.03, KiDS alone -0.45 ± 0.21. HSC reads +0.21 dex above KiDS, 3.8σ, 2.3σ with its own mass budget. The mass-dependence label comes from HSC alone, and we don't read it as one. A gas zero point of $+0.090 \pm 0.014$ on 5 X-COP clusters is unresolved. Total error is 25 % at 0.6 R_{500} and 38 % at R_{500}.

Can't decide. Temperature or gas as carrier of the deficit: NOT SEPARABLE, both footings, F 0.52, 0.60 (alt 0.53, 0.55) against a line of 3 (TG1). Binary-pulsar accelerations [12] on AQUAL against the round rule: NOT SEPARATED, $2\Delta \ln L$ −0.98 (alt −1.24) on 23 pulsars, with mock power 0; about 1200 (alt 940) would be needed (PA1).

Published AeST (UX2). This is the published theory [4], not our version. It fails rows 3 and 6 at every published configuration, Υ 0.411/0.377 and IC 1613 E −0.618/−0.655. That failure is weak, since both come from the MOND limit, and most other rows are CONDITIONAL or UNKNOWN. Its CMB fit holds at $g = 1$; at the physical g the benchmarks fail BAO and BBN. It reaches the CMB by carrying a fitted dust, ω_x 0.1206, linearly $w \approx 0$, $c_s \approx 0$. So at linear order it isn't a different answer from the retired sector. What that dust does in nonlinear wells nobody has computed, and that is the whole of row 4. One check failed, 2.16 against 0.5, voiding the lane's "how dust-like" scan.

Searching for an action (UX1, AS2, AS3). These results are empty for the listed candidates and the grammar, and neither is a no-go theorem. Of ten candidate classes, none structurally different passes; the one CONDITIONAL LEAD is v3s plus a separate dust, whose double count stands (capture 0.76). BIMOND [11] passes Q_2 with no screen, 0.029–0.107 of the line, and its tail still misses planets by 1.0e4 to 6.4e4 times their bounds. AS2 enumerated 2.17e11 raw combinations against analytic gates, no data: Tier A 0 of 717. 15 Tier B rows have no fail, and none is clean. The new corner it named, a khronon plus a scalar with $(D^2\chi)^2$, doesn't survive AS3: Q_2 and Ostrogradsky pass, but FRW growth fails, σ_8 ratio 0.0757 (alt 0.0817) against a line of 0.95, and so does the negative-charge instability.

Can one cold component do it (MC1, PC2)? On listed processes, MC1 ENDS AT REQUIREMENT 4. In budget, CMB and clusters plain dust is enough. Below 250 km s^{-1} it must be 66–82 % gone from discs with the measured response, and no computed process does that while it stays dust in the CMB. The required ratio to baryons is about zero in discs, 2–7 in clusters, 5.35 in the CMB. Density can't switch it: X-COP's need, $7.4 \times 10^{-4}\, M_\odot\, {\rm pc}^{-3}$, lies inside the disc range. A caution: a fixed-Υ pooled gate on record reads otherwise, the two gates disagree, and the verdict rests on our choice of gate (DW2 below traces that to the profile). A one-field phase change (PC2) is KILLED AT GATE 1 as a derivation, smallest miss −0.89 dex. With a counted mass of 3.74 eV it lands by fit, separates discs from clusters only marginally, +0.09 dex, and FRW thermalises. The only microphysics there that lands is particle-like, reported and not adopted.

That wall, and a max rule (DW2, MX1). These use a cold dark energy term as a test and reinstate nothing. How much must go depends on Υ: at 150–250 km s^{-1}, 53–79 % at 0.50, 0–39 % at the lowest measured values, where the frozen label is NOT DECIDED. So MC1's wall is CONDITIONAL on a disputed stellar mass; no row reads WALL OPENS, and with cold in the law's source it holds throughout. The gate disagreement is the profile: fixed-Υ rms reads 0.1025 on a profile holding about 5 times less mass and 0.1298 on the native one, line 0.110, and no process makes the former. Of 15 processes, none passes every gate; gravity-only rows miss by more than a factor of 2, and the nearest are particle-like. Separately, $g = \max(\text{law}, g_b + g_c)$ removes nothing and read 13–14 of 15 keys after two post-result changes. An independent freeze, not blind, FAILS: keys 13 of 15, G-OUT-M3 +3.80 (alt +3.28), joint p 0.017 (alt 0.028), Bullet subcluster M_{300} 1.15 against 2.3. X-COP's pass, rms 0.0964, is marginal, and it turned on a no-expansion clamp unstated in DW2's solver, found through MX1's missed check K2. The vector rule has no parent in the class tested. Its aligned parent keeps momentum and breaks weak equivalence between cold and baryons: galaxies drift 2.9–4.3 R_{200} from their cold in groups within 3 Gyr, and its own $f = 0$ cell fails X-COP, 0.119. Keeping universal fall makes the force one-way instead. MX1's map at the switch isn't converged (K7), and it has no action or stated reason.

Searching for a formula (BF1, BF2, SR3, BK2, AF1). None found, and none of this proves absence. BF1

level L3: 2.102e9 expressions, 0 Stage 1 survivors, as in a scrambled control, and a planted form recovered. Its level L4 is still running, with no result claimed. Its near-miss family, law $\times(\text{const} + \pi^2 g_{\rm gas}/a_0)$, fits X-COP out of sample and, on BF1's fixed-Υ statistic, breaks discs, rms 0.22–0.24 against a line of 0.175. AF1: NO FORMULA TRANSFERS. No clean variable carries the law's residual across samples, 0 CARRIES ACROSS on both footings, and SPARC has no survivor. Its frozen power plant failed and was replaced after the result.

BF2 took 3,952 hits from a snapshot of that running level and regraded them on FG1's free Υ, where 715 hold both rms and Υ (canonical), at twice or so the law's scatter in gas-rich slow galaxies. With no fitted float, two forms read NEAR MISS with 14 of 15 keys on both footings, as post-result picks, and a frozen-rule pick reads 15 of 15 and misses Υ at 0.47, 0.45. They earn no weight. The search is NOT ABOVE CONTROL: 19 real hits against 0, 15 and 266 for three scrambles on the same jobs, ratio to the mean 0.20. All three fail galaxy lensing, χ^2 845–885 on 56 against the law's 154 (canonical). All 24 present-gas Bullet cells read DEAD. No second implementation exists.

SR3, symbolic regression on the same free Υ with up to 2 fitted constants: NO CLEAN FORM, 0 of 141 real rows pass discs at the frozen line. A scrambled f reads 0.0554 against real 0.0567, and a pure cluster label passes every gate, so f acts as a sample label there. **Planted recovery failed** both frozen lines, so "nothing found" here isn't a proven absence, and any form on these fronts is a fit, not an identification. On the Bullet its graded forms have about the right mass and $\Delta\kappa$ wrong-signed at both subclusters, in all six rows.

BK2 enumerated dimensioned expressions, 2.7×10^{10} per footing at level L1: NO HIT on both footings, 0 through G1–G3 on the line or within errors. Its error reading of G3 is, by its own account, too strict, so the line is the fairer count. Closeness earns nothing: alt's scrambled control comes as close, 1.02 against 1.08 and 1.07 where 1 is on the line, and canonical's control hasn't finished. The closest no-fit row reads 15 of 15 keys on the one cell run. Its content is that total mass follows the gas, and it fails SPARC, dwarfs, IC 1613, DiskMass and the Bullet's sign. Level L2 is unfinished, temperature and dispersion aren't in the grammar, and check K3 failed on the IC 1613 row.

The a_0 formula: no statistical support that it is special (BK1, CG1). The record called the formula certified by a gate. That pass was a unit error. Repaired, it reads 5.51 bits at 5 %, under the 10-bit line, FDR-DEAD, and the relation doesn't land in that window on either footing. On a real power test at SPARC's 16.6 % half-width, 92.7 % of scrambled scalars decompose as simply as the formula, 0.11 bits. Among rivals on SPARC its coefficient is best in 1 of 6 cells, by 0.24 bits. A port of the search (BK1) tried 5.238e9 trees, with hits 1.03 times a scrambled control and 0 passes. This isn't a refutation. The test lacks power at this precision: 10 bits would need a half-width of 0.0032 %, about 5000 times sharper, so a true formula couldn't earn evidence today either. What's left is the old nearness of a_0 to cH, which rivals share, and the law's performance on data, which these lanes don't grade.

Routes to $\kappa = \frac{1}{2}$ (KF1). Four proposed routes: NOTHING FORCED. Two give no number. One gives 1/2, 1.447 or 1.772 by the length chosen, and only the length with no field equation behind it gives 1/2. One trades κ for $\kappa = 1/(2\xi)$. Forcing $\kappa = \frac{1}{2}$ is the same as forcing a length of exactly $c/\sqrt{G\rho}$. Five later routes, by hand algebra outside the frozen tests, also force nothing.

What the graded lines measure (DA1). VERDICT MOVES, PARTLY AUDITED, on lines only. Taken one at a time, other readings move cells of Table 2. With KiDS's bin $n(z)$ in $\Sigma_{\rm crit}$, C2's K reads 15 of 15. On a window of $0.5 \pm 25\,\%$, C1's S fails and the alt S of C3 and C4 passes; the 0.50 line's lower edge is the sources' centre, and no source read gives its 0.80. With SPARC's flow distances at native h, the S of C3 and C4 fails at 0.47, 0.48. Priors read as a mass scale put C1 (one background) and C3 inside on the Milky Way. X-COP's +0.111 mixes apertures: at one aperture it reads +0.092, 14 of 15 keys at f_* 0.034 and 15 at 0.0615, while the measured scale holds, at most +0.023, 12 of 15. Wall 1's f_Q and RM2's f aren't the same quantity; on RM2's own start the mass form reads 0.817 [0.454, 0.976], still above what RM2 needs. Each candidate still fails at least two lines on each reading, by inspection, since readings weren't run in combination except one pair. Several moves hurt the rule, and many inputs read CANNOT CHECK. These moves go to the owner.

A list of 96 rows (RQ3). A harness now grades any g on one list. Every candidate reads REJECTED. Known answers 110 of 110, after a first run at 103 of 110.

For the law, 49 rows compute from g, 8 are READ, 1 is OPEN DATA and 38 are NOT COMPUTABLE, mostly for want of an action or a Boltzmann run.

Table 4: RQ3 counts. They don't rank: rows are correlated in blocks, many are disputed, candidates compute different rows, and no cross-block covariance is on disk. Newton's 10 passes measure how little those rows ask.

candidate	footing	pass	fail	form-dep.	computed	keys
Newton	both	10	38	0	48	4 of 15
a_0 law	canonical	22	25	2	49	10
a_0 law	alt	20	27	2	49	10
UX3 term, k fit	canonical	29	15	0	44	14
UX3 term, k fit	alt	25	19	0	44	10
cold halo, known-answer test	both	5	6	0	11	not computable

8 What a successor must do

Table 5 lists twelve requirements.

Table 5: Requirements from combining the failures. Older rows are on older chains, and FG1 supersedes any it covers. Coverage: an inventory that opened about 90 of about 330 lanes.

#	must	law	native halos
1	keep galaxies tight	**0.0541**	0.0682 / 0.0694
2	stay flat with surface brightness	−0.016	−0.173
3	sit on Υ at every speed	0.52, alt 0.44	0.75, running 1.20 to 0.57
4	show little net contraction in spirals	not needed	needs $f \leq 0.2$, measured 0.872; beside the law, 66–82 % gone (MC1), conditional on Υ (DW2)
5	fit dwarfs	−0.025; ultra-faints still +5.08 (UX4)	+0.138; 0 of 5 classes
6	fit IC 1613	−0.593	−0.474
7	fit the Milky Way	AQUAL, QUMOND INFEASIBLE; round rule only	asks 4e10 $M_\odot$ inside R_0
8	fit isolated lenses	shape preferred	+366.0
9	add mass in cluster cores	A 1.91	G-IN −0.49, G-MID +2.17
10	match outer cluster shear	G-OUT-M3 +4.76; slope 0.47 against 0.90	G-OUT-M3 +4.41
11	act as dust in the CMB	no cold term	$z_{\rm on} \in [3 \times 10^4, 10^6]$; ω_x 0.1197 (UX2)
12	have an action	open; 0 of 717 in AS2's Tier A	$P(X)$ dust limit only

The collisions. Rows 4 and 11: dust in the CMB falls into galaxies. Rows 1–3 and 10: the amount needed runs from about 0.8 of native on SPARC to 1.22–1.80 on clusters. Rows 7 and 8: systems of the same mass want different things. Selectors closed on inputs tested: potential, σ, acceleration, environment, formation epoch, gas state, bars. None of this is a theorem.

One family left outside. A 5–11 eV fermion would address rows 4 and 11 directly. Its forest kill isn't on the current footing: a 1-D proxy against a ΛCDM yardstick falls 1000–5000 times short, undecided on raw forest data. Other liabilities stand, and it wouldn't fix rows 6 or 7. It's outside the no-particle rule.

9 Dated tests, and their real power

- **ALMA U4__28473, public 2026-11-06 02:42:31 UTC** (PAPER66 v1.2, DOI 10.5281/zenodo.23868979). It tests this galaxy's mass budget, not $a_0(z)$, and one disc excludes no model.
- **GS4__24110,** [CI](2–1), 2026-11-28 by earlier metadata, to be rechecked. Every $n = 1$ against $n = 2$ pair is at or below 0.51σ, so it can't separate them.
- **Gaia DR4, about December 2026** (pre-registered 2026-07-16, with later amendments). Wide binaries:

Newton 1.000, an ungated MI band 1.05–1.10, AQUAL point 1.137, MOND's benchmark 1.33; the DR3 dry run's 1.205 ± 0.035 is evidence for nothing. An amendment registers a gated branch for which 1.000 inside 30 kAU is also predicted, so that branch needs its own test beyond 50 kAU, about 2000 clean pairs per bin. The solar-acceleration dipole's kill bands are void under the amended theory, so that front returns no verdict. *Reading, not a frozen line:* these bands were written for the action-carrying framework. The bare law here contains no gate, so a Newtonian result at 2–30 kAU would count against it.

- **SP1, SP2 sealed lensing predictions.** Scored as frozen, including the retired route's rows.
- **Named, not scheduled:** one clean deep-MOND a_0 point at $z \approx 2.5$ to about 0.10 dex. That's the measurement that would actually test the dark energy tie.
- **Named, not run:** a per-galaxy MUSE-DARK refit with this law imposed and a stated baryon-mass error budget. If $a_0(z \sim 1)$ stays well above the band with mass systematics under control, the band fails.

10 Method

Each lane commits frozen criteria alone before any script or number, runs a MUTATE twin that must fail, reruns byte-identically and publishes failures under frozen labels. Most lanes were not blind. Grading was done by the same programme that proposed the models; the clean-room regrade tested code, not independence of definitions. No outside group has reproduced these results. Lanes in Section 7 were run by AI subagents after the retirement, and their failed checks are listed in each README.

Not claimed

A complete, closed or unified theory. That the data favour this framework. κ, T or any abundance as derived. That the tie of a_0 to dark energy originated here. That the requirements list is a no-go theorem.

Data and code

All lanes named here are in `rebuild_2026_10_22/` and `campaign_fresh_gravity/`. Scorecard: `FG1_final_grade`. Law's scorecard: `LW1_law_scorecard`. Ledger, still a working draft: `CLOSING_LEDGER_2026-11-01_DRAFT.md`. Lanes new in v1.1 are in `rebuild_2026_10_22/`, under the names in Section 7: `AR1_action_response`, `UX3_cluster_gas_term`, `CU1_gas_term_replication`, `CG1_a0_certificate`, `RQ3_acceptance_v2`, `BK2_dimensional_law_search`, `DA1_definition_audit` and others. BF1 and BK2 are still running, and their READMEs are the record of what has landed. Audit: `PAPER67_audit.py`.

References

[1] M. Milgrom, ApJ 270, 365 (1983).

[2] M. Milgrom, Phys. Lett. A 253, 273 (1999).

[3] C. Limbach, D. Psaltis, F. Özel, arXiv:0809.2790 (2008).

[4] C. Skordis, T. Złośnik, arXiv:2007.00082 (2020).

[5] L. Blanchet, C. Skordis, arXiv:2404.06584 (2024).

[6] A. A. Dutton et al., arXiv:0801.1505 (2008).

[7] S. Aniyan et al., MNRAS 456, 1484 (2016), arXiv:1511.06047.

[8] S. Aniyan et al., MNRAS 476, 1909 (2018), arXiv:1802.00465.

[9] S. Aniyan et al., arXiv:2010.03991 (2020).

[10] Clowe et al., arXiv:astro-ph/0608407 (2006).

[11] M. Milgrom, arXiv:0912.0790 (2009).

[12] Donlon et al., arXiv:2501.03409 (2025).

[13] B. I. Ciocan et al., A&A 709, L16 (2026), arXiv:2604.22613.

[14] G. T. Alcock, MNRAS 551, L1 (2026), arXiv:2609.38796.

[15] Jeanneau et al., A&A 709, A120 (2026), arXiv:2603.28856.